Wavelength Dispersion of Ti Induced Refractive Index Change in $LiNbO_3$ as a Function of Diffusion Parameters

S. FOUCHET, A. CARENCO, C. DAGUET, R. GUGLIELMI, AND L. RIVIERE

***Abstract*—*Z*-cut *X*-propagating, $LiNbO_3$: Ti planar waveguides have been characterized over a wide range of diffusion conditions. Measurements of Ti concentration and refractive indexes have revealed Gaussian profiles. Their dependence with diffusion conditions have been studied. The relation, established at several wavelengths ($\lambda = 0.6328$ μm, 1.1523 μm, 1.523 μm), between extraordinary and ordinary refractive index change $\Delta n_{e,o}(z)$ and titanium concentration $C(z)$ is found to be nonlinear and of the form:**

$$\Delta n_{e,o}(z) = A_{e,o}(C_o, \lambda)\,(C(z))^{\alpha_{e,o}}.$$

While $\alpha_{e,o}$ is a constant, the proportionality coefficient $A_{e,o}$ depends not only on the wavelength λ, but also on the diffusion conditions, characterized by C_o, the Ti surface concentration of the diffused guide. Such a nonlinearity of Ti diffusion in $LiNbO_3$ has been revealed thanks to a complete and wide range investigation on both indiffused Ti and index profiles.

I. Introduction

THE INTEREST of $LiNbO_3$: Ti waveguides in integrated optics is no more to be justified. To take full profit of their low transmission loss and good electrooptic properties, it is fundamental to have a precise knowledge of the refractive index profile induced by titanium indiffusion in the $LiNbO_3$ substrate as a function of fabrication parameters, at the considered wavelength. In particular, the 1.3–1.6 μm spectral range is of high interest for optical communication systems. To date, work to understand the mechanisms of titanium diffusion in $LiNbO_3$ is in progress [1], but up to now, the origin of Titanium induced refractive index change has not been satisfactorily explained [2]. As a consequence, these profiles have to be experimentally determined.

Several papers have already dealt with that problem (see for example [3]–[6]). On one hand, the diffusion conditions—air, argon, or oxygen atmosphere, either wet or dry—are very different from one laboratory to the other [7]; consequently the dispersion of reported results is very important. The disagreement between data, even for diffusion conditions apparently identical, could be due to a poor control of diffusion process (Li out-diffusion for example) or crystal quality. Despite these differences, most of published papers agree to consider that Δn_e (extraordinary index change) and C (titanium concentration) are proportional, whereas Δn_o (ordinary index change) has a nonlinear behavior versus C [3], [6], [8]. But these tendencies have to be confirmed by investigations on waveguides made in a wider range of diffusion conditions.

On the other hand, the wavelength dependence of refractive index change has been explored only up to 1.15 μm [6], [9]. For 1.3- or 1.5-μm applications, the wavelength dispersion is not taken into account, or when it is, the effect is adjusted afterwards to the experimental characteristics of the device [10].

The purpose of this paper is to study:

a) the indiffused titanium concentration profiles (Section III),
b) the refractive index profiles at different wavelengths (0.6328 μm–1.1523 μm–1.523 μm) (Section IV), and
c) the way they are related over a wide range of diffusion parameters (Section V).

II. Samples Preparation

The $LiNbO_3$ crystal of optical grade quality corresponds to the congruent composition [11]. The 16 wafers used here are $-c$-cut from two different boules. The titanium film deposited by sputtering (thickness $\tau = 300$–2100 Å) is diffused in a flowing atmosphere of dry argon, between $t = 9$ h and 96 h, either at $T = 1000°$C or 1050°C, (Fig. 1) and then cooled in oxygen. The fabricated waveguides exhibit a wide range of titanium surface concentrations (from 0.2 percent to 2 percent in mass) and a wide range of diffusion depth (from 4 to 15 μm) as it will be developed further. One third of the surface remains Ti free, in order to check after diffusion the stability of the substrate [12]. When using the optical technique described in [13], no index variation (either positive or negative) is observed on that part of the substrate. Only the samples diffused at 1050°C during 96 h revealed a negligeable decrease of the extraordinary index ($\Delta n_e \sim -7 \times 10^{-5}$), which could be attributed to a small amount of Li in-diffusion. It is important to note that all the other samples are stable within the accuracy of the control technique.

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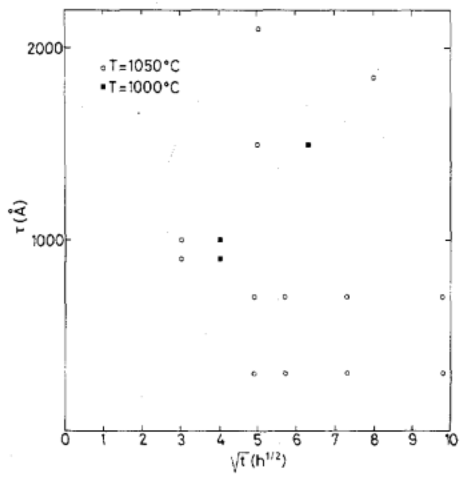

Fig. 1. Plot of the various samples (titanium film of thickness τ(Å), diffused t(h) at T(°C)).

III. Microprobe Analysis

The Titanium concentration of diffused waveguides is determined with an electron microprobe (CAMEBAX). The $K\alpha$ line is observed under a 15-kV voltage, and the "trace" option of the correction program is used for data processing. According to the experimental conditions, the elementary volume which is analyzed, roughly corresponds to 0.5 μm^3.

A scanning of the surface along Y (Fig. 2), across the limit ($Y = Yo$) of the diffused area, gives the lateral diffusion profile of titanium. To determine the profile along Z, a 1° bevel is polished on the sample (Fig. 2).

A. Theoretical Approach

The diffusion of titanium in lithium niobate in the assumption of no external perturbation, can be described by Fick's equation [14]. As the problem here has only to be solved in the Y, Z plane (Fig. 2):

$$\frac{\partial C}{\partial t} = D_y \frac{\partial^2 C}{\partial Y^2} + D_z \frac{\partial^2 C}{\partial Z^2} \tag{1}$$

where C is the Ti repartition in the Y, Z plane: $C(Y, Z)$, and D_y and D_z, are the diffusion coefficients along the crystal axes Y and Z, respectively, and are supposed independent of C. D_y and D_z follow an Arrhenius law:

$$D_{y,z} = D_o^{y,z}\, e^{-E_o^{y,z}/kT} \tag{2}$$

where

D_o^y, D_o^z	diffusion constants,
E_o^y, E_o^z	activation energies in the Y and Z directions,
k	Boltzmann's constant, and
T	diffusion temperature (°K).

Using twice a Fourier transformation, as the diffusion takes place in a semi-infinite medium, (1) becomes:

$$C(Y, Z) = C_I(Y, Z) * \frac{1}{\pi} \frac{1}{\sqrt{2D_z t}}\, e^{-Z^2/4D_z t}\, \frac{1}{\sqrt{2D_y t}}\, e^{-Y^2/4D_y t} \tag{3}$$

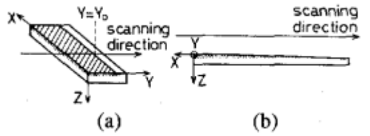

Fig. 2. Z-cut X-propagating $LiNbO_3$: Ti planar waveguides. The sample is partially covered with Ti(2/3 of the surface). In (a) microprobe scanning of the surface along the Y axis determines the titanium indiffused lateral profile. In (b) the depth profile is analyzed on a ~1° bevelled surface.

where

t	diffusion time, and
$C_I(Y, Z)$	initial titanium concentration just before diffusion ($t = 0$).

Let assume first, that the titanium film of thickness τ and concentration C_f is infinite in the Y and X directions. In (3) $C_I(Y, Z)$ does not correspond to the actual initial repartition of Ti, since (1) does not describe the diffusion through interfaces. At $t = 0$, (1) supposes that titanium is embedded in the crystal. That leads to define an equivalent thickness τ' of the initial Ti film, taking into account the saturation limit of Ti concentration in $LiNbO_3$ host (C_s):

$$C_f \tau = C_s \tau' \tag{4}$$

which expresses the conservation of the total quantity of titanium. Then

$$C_I(Y, Z) = C_s, \quad \text{when } -\tau' \le Z \le 0$$
$$= 0, \quad \text{elsewhere}$$

From (3), it becomes

$$C(Y, Z) = C_s \left(\operatorname{erf}\left(\frac{Z + \tau'}{d_z}\right) - \operatorname{erf}\left(\frac{Z}{d_z}\right) \right) \tag{5}$$

where $d_z = 2\sqrt{D_z t}$ is the diffusion depth.

It can be noted: first, in the case of a very deep diffusion ($\tau'/d_z << 1$), the profile can be approximated by a Gaussian function

$$C(Y, Z) = \frac{2}{\sqrt{\pi}} C_f \frac{\tau}{d_z} e^{-Z^2/d_z^2}. \tag{6}$$

Second, as long as the titanium film is not completely diffused, τ' can be considered as infinite, and the profile is a complementary error function:

$$C(Y, Z) = C_s \operatorname{erfc}(Z/d_z). \tag{7}$$

If the Ti film to be diffused is now semi-infinite and limited to $Y = Yo$ (Fig. 2), it becomes:

$$C_I(Y, Z) = C_s, \quad \text{when } -\tau' < Z < 0 \text{ and } Y \le Y_o$$

$$C_I(Y, Z) = 0, \quad \text{elsewhere}$$

and

$$C(Y, Z) = \frac{C_s}{2} \left(\operatorname{erf}\left(\frac{Z + \tau'}{d_z}\right) - \operatorname{erf}\left(\frac{Z}{d_z}\right) \right) \operatorname{erfc}\left(\frac{Y - Y_o}{d_y}\right) \tag{8}$$

with $d_y = 2\sqrt{D_y t}$.

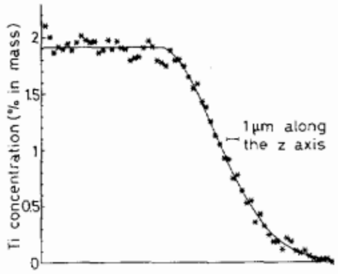

Fig. 3. Ti concentration depth profile of a Z-cut $LiNbO_3$: Ti sample (τ = 1500 Å, t = 40 h, T = 1000°C). The measurement pitch corresponds to about 0.35 μm along Z. The solid curve is fitted on the experimental data (C_o = 1.9 percent, d_z = 6.2 μm).

At the surface of the sample ($Z = 0$)

$$C(Y) = \frac{C_s}{2} \operatorname{erf}\left(\frac{\tau'}{d_z}\right) \operatorname{erfc}\left(\frac{Y - Y_o}{d_y}\right). \tag{9}$$

So a complementary error function is expected from the surface analysis of Ti concentration.

B. Experimental Results

1. Depth Profile: The microprobe scanning along the polished bevel with an effective pitch along Z of 0.35 μm leads to a set of Ti concentration values which yield a Gaussian function with a very good accuracy (Fig. 3) for the various samples. The two unknown parameters, the Ti surface concentration C_o, and the diffusion depth d_z of the Gaussian distribution of Ti:

$$C(Z) = C_o\, e^{-Z^2/d_z^2} \tag{10}$$

are determined by the least squares method. According to (6) a Gaussian profile should have parameters verifying:

$$C_o = \frac{2}{\sqrt{\pi}} C_f \frac{\tau}{d_z} \tag{11a}$$

$$d_z = 2\sqrt{D_z t},\ D_z = D_o^z\, e^{-E_o^z/kT}. \tag{11b}$$

The set of fitted parameters C_o, d_z yield fairly well (11a) and (11b).

a. C_o is proportional to τ / d_z (Fig. 4): The data are fitted with a rather good accuracy by a straight line

$$C_o = 0.77\ \tau/d_z. \tag{12}$$

According to (11a), it can be calculated

$$C_f = 3.8 \times 10^{28} \quad \text{atom}/\text{m}^3$$

figure which can be favorably compared with the atomic density of solid Ti : 5.6 × 10^{28} atom/m^3.

b. d_z and $\sqrt{t}$ are proportional (Fig. 5): both at 1000°C and 1050°C. The coefficients E_o^z and D_o^z of the Arrhenius law (2) are deduced from the two slopes:

$$\begin{aligned} E_o^z &= 2.60 \text{ eV} \\ D_o^z &= 5.0 \times 10^9\ \mu\text{m}^2/\text{h}. \end{aligned} \tag{13}$$

It is to be noticed that the Ti profile resolution is limited by the microprobe pitch (~0.35 μm); smaller variations in the profile, such as bumps or deeps which could exist at the surface [15], [16] are impossible to detect here.

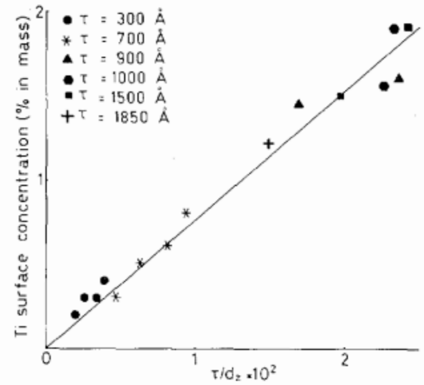

Fig. 4. Measured mass concentration of Ti at the surface of the samples, C_o, versus τ / d_z. The solid line is fitted.

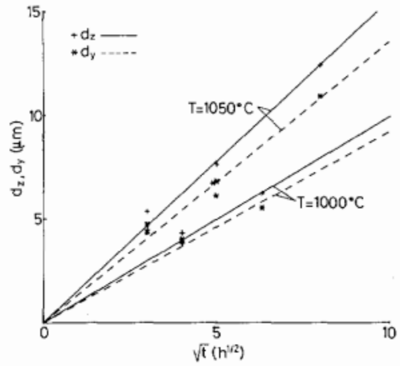

Fig. 5. Experimental values of diffusion lengths along the different crystal axes (d_y, d_z) versus time for different temperatures, and fitted straight lines.

2. Lateral Profile: Surface analysis of the diffused samples gives the lateral profile of titanium concentration (Fig. 2). As expected from (9) experimental results fit very well a complementary error function:

$$C(Y) = C_o' \operatorname{erfc}\left(\frac{Y - Y_o}{d_y}\right) \tag{14}$$

whose parameters C_o', Y_o, and d_y are determined by the least squares method. The linear dependence of d_y versus $\sqrt{t}$ at both temperatures (Fig. 5) leads to the following Arrhenius coefficients:

$$\begin{aligned} E_o^y &= 2.22 \quad \text{eV} \\ D_o^y &= 1.35 \times 10^8 \quad \mu\text{m}^2/\text{h}. \end{aligned} \tag{15}$$

As a means to control the validity of this model, stripe waveguides of different widths (from 3 to 10 μm) are diffused in the same conditions (900 Å-1050°C-16 h). The microprobe measurements give an average value:

$$d_y = 5.45 \quad \mu\text{m}$$

which is in perfect agreement with the value calculated from (15)

$$d_y = 5.42 \quad \mu\text{m}.$$

TABLE I
COMPARISON OF PUBLISHED RESULTS ON E_o, D_o, $D = D_o e^{-E_o/kT}$, FOR THE TITANIUM CONCENTRATION PROFILE ALONG THE Y AXIS, FOR Y AND Z CUT $LiNbO_3$

Reference	This work	[17]	[6]	[16]	[18]	[19]	[2]	[5]
$LiNbO_3$ cut	Z	Z	Z	Z	Y	Y	Y	Y
Diffusion Atmosphere	$Ar_{(+)}$dry	Air dry	O_2 (*)	Ar (+)	Air dry (+)	O_2wet	$Air_{(*)}$	Ar(+)
Outdiffusion	No	Yes		Yes	Yes			Yes
E_o(eV)	2.22	1.48			2.26	2.22	2.18	
D_o (μm^2/h)	$1.35\ 10^8$	$2.65\ 10^5$			$1.5\ 10^8$	$1.09\ 10^8$	$7.9\ 10^7$	
D, T=1000°C (μm^2/h)	0.214	0.366	0.396	0.349	0.169	0.177	0.184	0.166
D, T=1050°C (μm^2/h)	0.460	0.610			0.369	0.382	0.391	

(+) When Ar is used as a diffusion atmosphere, the samples are then cooled in Oxygen.

(*) Diffusion of a TiO_2 film

TABLE II
TI DIFFUSED PLANAR WAVEGUIDES ON Z-CUT $LiNbO_3$—COMPARISON OF PUBLISHED RESULTS FOR THE INDIFFUSED TITANIUM CONCENTRATION PROFILE (Ti), AND FOR INDUCED EXTRAORDINARY (E) OR ORDINARY (O) INDEX PROFILES

References		This work	[17]	[19]	[20]	[21]	[22]	[23]		[6]	[10]	[24]	[16]	[25]
Diff. Atmosphere		$Ar_{(+)}$dry	Air dry	O_2wet	wet	$Ar_{(+)}$dry	O_2wet	O_2wet	O_2 dry	O_2(*)	O_2wet	Air	$Ar_{(+)}$	$Ar_{(+)}$wet
Out		No	Yes			No				No		Yes	Yes	
Eo (eV)	Ti	2.60	2.95	2.22										
	E	2.77			2.70	1.98	1.84							
	O	2.70					1.12							
D_{o} (μm^2/h)	Ti	$5.0\ 10^9$	$1.4\ 10^{11}$	$1.09\ 10^8$										
	E	$2.6\ 10^{10}$			$8.3\ 10^9$	$1.1\ 10^7$	$1.21\ 10^4$							
	O	$2.2\ 10^{10}$					$3.77\ 10^6$							
D (μm^2/h) T=1000°C	Ti	0.247	0.270	0.177										
	E	0.282			0.174	0.157	0.209			0.612				
	O	0.447					0.446			0.612		0.286	0.504	
D (μm^2/h) T=1050°C	Ti	0.606	0.792	0.382							0.360			
	E	0.733			0.441	0.310	0.393	1.12	0.684					
	O	1.13					0.657	0.857	0.958					0.432

(+) When Ar is used as a diffusion atmosphere, the samples are then cooled in Oxygen.

(*) Diffusion of a TiO_2 film.

3. Discussion: When comparing Ti diffusion data to the literature (Tables I and II) where dispersion of results is quite important, it appears that our figures are in the range of published data.

The values of Dy (deduced from lateral diffusion measurements on Z-cut crystals) are compatible with published results, and are close to figures concerning Y-cut crystals (Table I: [2], [18], [19]).

TABLE III
$LiNbO_3$ REFRACTIVE INDEXES

Reference	λ = 0.6328μm		λ = 1.1523μm		λ = 1.523μm	
	n_o	n_e	n_o	n_e	n_o	n_e
This work	2.2865 ± 2 10-4	2.2020 ± 2 10^{-4}	2.2273 ± 4 10^{-4}	2.1512 ± 4 10^{-4}	2.2125 ± 4 10^{-4}	2.1383 ± 4 10^{-4}
|26|	2.2865	2.2024	2.2273	2.1515	2.2121	2.1388
|11|	2.2869 ± 2 10^{-4}	2.2022 ± 2 10^{-4}				

To our knowledge, only two complete investigations of Ti diffusion along the Z axis are available [17], [19]. Ti diffusion depths (Table II) are in good agreement with Fukuma *et al.*'s results [17]. But there is a discrepancy with Holmes *et al.*'s results [19] on the Z axis. It turns out that in our case, Ti diffusion cannot be considered as isotropic.

IV. Optical Characterization

The optical characterization consists in establishing the refractive index profile from the guided mode spectrum analysis.

A. Experimental Approach

First, the refractive indices of the substrate and of rutile prisms are determined by the prism minimum deviation method at the three wavelengths. Two different boules of $LiNbO_3$ are used; their indices are checked to be the same within the experimental accuracy, corresponding to the congruent composition [11]. Data are gathered in Table III. Both indices values agree very well with the Smith *et al.* results in [26].

Secondly, the effective index spectra of the guided modes are measured using the prism coupling method [27], with the help of detectors and infrared camera.

Once the effective indices of the waveguide are known, the problem is to recover the entire refractive index distribution. There, the inverse WKB method, with and without additional assumption on the analytical index profile, is used for the computation.

a) Without any assumption [28], the profiles are very close to Gaussian functions, even for the waveguide exhibiting the highest surface concentration ($C_o \sim 2$ percent) as illustrated in Fig. 6.

b) a Gaussian profile hypothesis leads to a set of effective indices in good agreement with the experimental one.

The extraordinary and ordinary refractive index changes ($\Delta n_{e,o}$, respectively) of planar waveguides, are described by

$$\Delta n_{e,o}(Z) = \Delta n^s_{e,o}\, e^{-Z^2/d^2_{e,o}} \qquad (16)$$

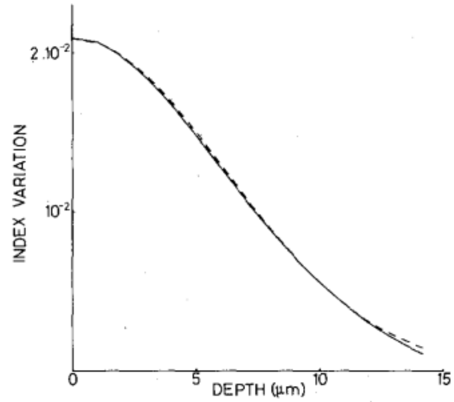

Fig. 6. Extraordinary index profile of the $LiNbO_3$:Ti sample ($\tau = 2100$ Å, $t = 25$ h, $T = 1050°C$) exhibiting the highest Ti concentration at the surface ($C_o = 2$ percent). The solid line curve is the profile calculated without any analytical assumption. The dashed line curve is a Gaussian function. The maximum index difference between the two curves is about 4×10^{-4}.

with $\Delta n^s_{e,o}$ the maximum value, and $d_{e,o}$ the depth of refractive index change profiles (respectively, extraordinary and ordinary).

B. Results

The experimental results show that d_e and d_o are independent of τ and of λ, the optical wavelength. They are only functions of the diffusion parameters t and T. d_e and d_o are plotted versus $\sqrt{t}$ at $T = 1000°C$ and $1050°C$ (Fig. 7). A linear dependence describes their behavior. It appears that $d_z < d_e < d_o$. Arrhenius coefficients can be calculated as follows.

For the extraordinary index

$$E^e_o = 2.77 \quad \text{eV}$$
$$D^e_o = 2.6 \times 10^{10} \quad \mu m^2/h. \qquad (17)$$

For the ordinary index

$$E^o_o = 2.70 \quad \text{eV}$$
$$D^o_o = 2.2 \times 10^{10} \quad \mu m^2/h. \qquad (18)$$

It is worthwhile to note that the activation energies for n_e and n_o are very identical, and also very close to the

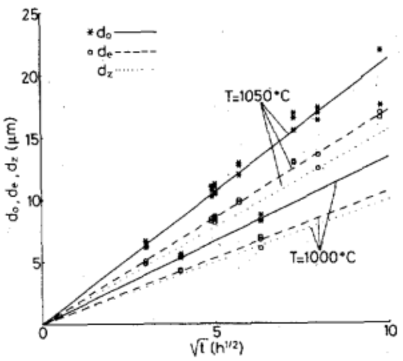

Fig. 7. Plots of widths of ordinary and extraordinary Gaussian refractive index profiles (respectively d_e and d_o) versus time as in Fig. 5. The dotted line corresponds to previously fitted d_z curve.

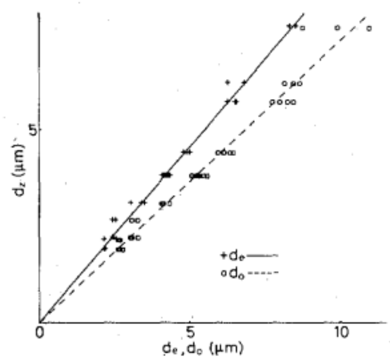

Fig. 8. Diffusion depth d_z, versus widths of the extraordinary and ordinary Gaussian refractive index profiles (respectively, d_e and d_o) at different wavelengths and corresponding fitted lines (solid and dashed, respectively).

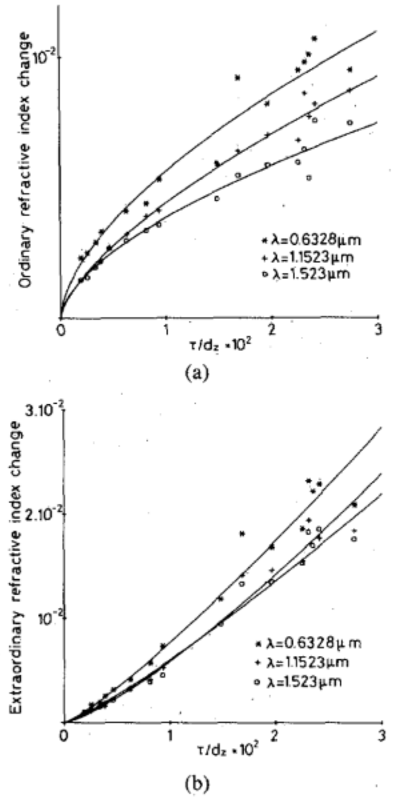

Fig. 9. (a) Maximum ordinary (Δn_o^s) and (b) extraordinary (Δn_e^s) refractive index changes as a function of τ / d_z at different wavelengths. The experimental data are fitted with power laws.

activation energy of titanium diffusion (2.60 eV), as it was reasonably expected.

These results agree very well with previously published figures as can be seen on Table II.

V. Relation Between Refractive Index Change and Ti Concentration

A. Analytical Relations

The problem now, is to establish the direct relation between $\Delta n_{e,o}$ and C, i.e., to mathematically define it from the analytical expressions found, respectively, in Sections IV and III.

It has been previously demonstrated that both extraordinary and ordinary refractive index and titanium profiles are Gaussian functions. So they are related by an expression of the form

$$\Delta n(Z) = A\big(C(Z)\big)^{\alpha} \tag{19}$$

where A and α are independent of the Z coordinate. It comes from (6) and (16) that

$$\Delta n^s_{e,o} = A_{e,o}\left(C_f \frac{2}{\sqrt{\pi}}\right)^{\alpha_{e,o}} \left(\frac{\tau}{d_z}\right)^{\alpha_{e,o}} \tag{20a}$$

$$d_{e,o} = d_z / \sqrt{\alpha_{e,o}}. \tag{20b}$$

At a given wavelength, the relation between Ti concentration and induced refractive index change (extraordinary and ordinary, respectively) will be a power law with constant coefficients ($A_{e,o}$ and $\alpha_{e,o}$), only if both (20a) and (20b) are simultaneously verified. $A_{e,o}$ and $\alpha_{e,o}$ are to be determined from the experimental data.

First, the plot, versus d_z, of d_e and d_o, obtained at the various λ, lets a linear dependence appear (Fig. 8). From the slope of each straight line, it is calculated:

$$\begin{aligned} \alpha_e &= 0.83 \\ \alpha_o &= 0.53. \end{aligned} \tag{21a}$$

Secondly, $\Delta n^s_{e,o}$ versus τ / d_z, plotted on Fig. 9, follows with a good precision, a power law as expected from (20a). The least squares method used to fit the data leads to another evaluation of α_e and α_o:

$$\begin{aligned} &\text{for } \lambda = 0.6328\ \mu\text{m}, \quad \alpha_o = 0.56, \quad \alpha_e = 1.2 \\ &\text{for } \lambda = 1.1523\ \mu\text{m}, \quad \alpha_o = 0.45, \quad \alpha_e = 1.3 \\ &\text{for } \lambda = 1.523\ \mu\text{m}, \quad \alpha_o = 0.39, \quad \alpha_e = 1.2. \end{aligned} \tag{21b}$$

The wavelength dependence found in (21b) (particularly noticeable for α_o), and the obvious discrepancy of the figures in (21a) and (21b) for any wavelength, is a proof that $A_{e,o}$ is not only dependent on λ, but also on the diffusion conditions. So α_e and α_o are completely determined by (21a), and then $A_{e,o}$ has to be adjusted so that the curves defined in (20a) fit the experimental data. Since a given value of C induces increasing values of Δn_e and

Δn_o as the surface concentration of the sample C_o increases, the proportionality coefficient $A_{e,o}$ depends on C_o i.e., τ/d_z according to (12). This dependence is appreciable for the extraordinary index and smaller for the ordinary one.

Using a series expansion, limited to the first order, it is possible to transform (20a) into

$$\Delta n_{e,o}^s = \left(B_0(\lambda) + B_1(\lambda) \cdot \tau/d_z\right) (\tau/d_z)^{\alpha_{e,o}}. \quad (22)$$

For each λ, the experimental data are fitted to (22) using (21a) $\alpha_e \simeq 0.8$ and $\alpha_o \simeq 0.5$ (Fig. 10).

In the range $0.6 \le \lambda(\mu m) \le 1.6$, the dispersion of B_0 at B_1 can be approximated by second-order polynomial expressions:

$$\begin{aligned} n_o \quad & B_0 = 6.53 \times 10^{-2} - 3.15 \times 10^{-2}\lambda + 7.09 \times 10^{-3}\lambda^2 \\ & B_1 = 0.478 + 0.464\lambda - 0.348\lambda^2 \\ n_e \quad & B_0 = 0.385 - 0.430\lambda + 0.171\lambda^2 \\ & B_1 = 9.13 + 3.85\lambda - 2.49\lambda^2. \end{aligned} \quad (23)$$

The simultaneous verification of (20a) and (20b) validates the power law relation between Gaussian expressions of the index profiles and of titanium concentration. Using (11a), it is seen that

$$\Delta n_{e,o}(Z) = A_{e,o}(\lambda, C_o)\left(C(Z)\right)^{\alpha_{e,o}}$$

$$A_{e,o}(\lambda, C_o) = \left(\frac{2}{\sqrt{\pi}} C_f\right)^{-\alpha_{e,o}} \left(B_0(\lambda) + B_1(\lambda) \cdot \frac{\sqrt{\pi}}{2C_f} \cdot C_o\right). \quad (24)$$

B. Discussion

The relation between ordinary or extraordinary refractive index change and titanium concentration is a power law, given in (24), whose proportionality coefficient depends on λ and on the diffusion conditions (i.e., on C_o which is the Ti surface concentration of the diffused samples). Both (23) and (24) characterize Ti diffused waveguides between $\lambda = 0.6$ μm and $\lambda = 1.6$ μm, as long as the Ti indiffused concentration profile along the Z axis is a Gaussian function. The problems to be solved now would be:

First, to find the limit of the Gaussian model validity, i.e., to determine the maximum τ/d_z value, which leads to refractive index and Ti concentration profiles with a Gaussian shape.

Secondly, to test the validity of (24) for shallower diffusion, i.e., increasing τ/d_z.

The above relation given in (24) can be considered as nonlinear from two points of view:

1) the power law, and
2) the dependence of the coefficients on the diffusion conditions, i.e., C_o.

The first aspect of the nonlinearity has already been noticed in the literature. Minakata *et al.* [3], Ludtke *et al.* [4], Vollmer *et al.* [5], and C. Bulmer *et al.* [22] have found that Δn_o has a sublinear behavior versus C. Some

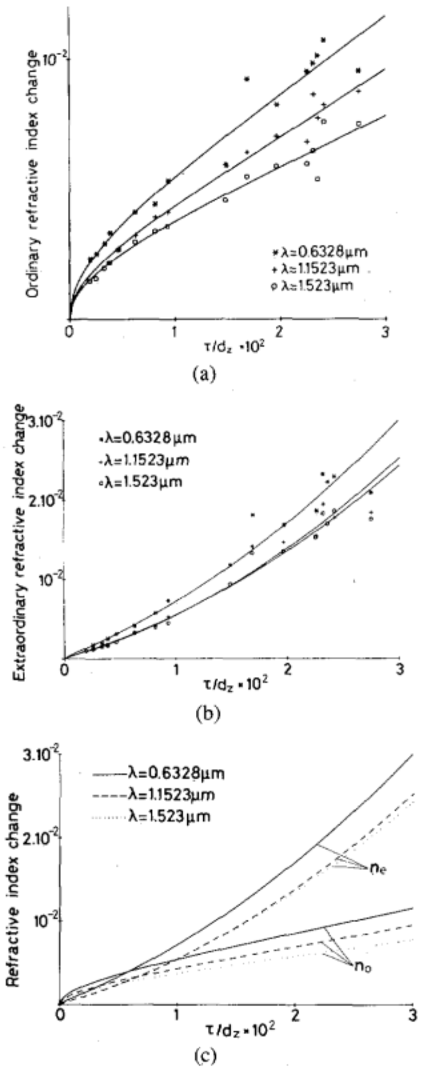

Fig. 10. (a) Experimental data of Δn_o^s and (b) Δn_e^s versus τ/d_z, at different wavelengths. The analytical expressions of the fitted curves are of the form $\Delta n_{e,o}^s = (B_0(\lambda) + B_1(\lambda) \cdot \tau/d_z) \cdot (\tau/d_z)^{\alpha_{e,o}}$ with $\alpha_e \sim 0.8$ and $\alpha_o \sim 0.5$. (c) Illustration of the relative position of the previous curves.

power laws have already been published. At $\lambda = 0.6328$ μm, Vollmer *et al.* [5], and Ludtke *et al.* [4] found $\alpha_o = 0.62$ and 0.55, respectively, figures close to our (21b) $\alpha_o = 0.56$. In most cases, Δn_e is reported as being proportional to C [3], [5], [6]. This is due to the fact that the curves Δn_e versus τ/d_z are quite close to straight lines, especially when limiting the experimental range of τ/d_z. Only, in [5], Δn_e follows a 0.88 power dependence, value of α_e which is very close to ours, in spite of the presence of out-diffusion in their experiments. The straight line fitting the $(\tau/d_z, \Delta n_e)$ experimental points in [22] does not cross the origin of axes; a curve having the analytical behavior described previously would be in best agreement.

As far as the second aspect of nonlinearity is concerned, it has never been found. Both relations (20a) and (20b) are to be considered to be able to deduce a relation such as (19). The experimental investigation of these two relations is the key point to discover that $A_{e,o}$ is not only λ dependent but depends also on C_o. This means that the

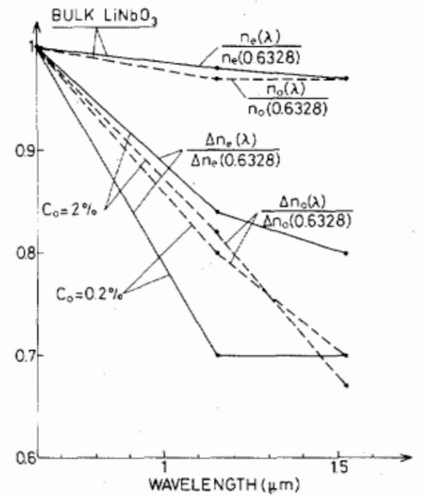

Fig. 11. Dispersion of refractive index change induced by Ti indiffusion compared to that of bulk refractive index. The different quantities are normalized to 1 at $\lambda = 0.6328$ μm.

hypothesis of a local relationship between $\Delta n_{e,o}$ and C is not verified. A given value of titanium concentration in the crystal induces increasing values of refractive index change, as the surface concentration C_o increases. The physical explanation remains to be found. However, photoelasticity could be considered as largely responsible of the refractive index change [2]. Ti diffusion generates marked defects in the crystal. Recent work has revealed that in the first part of the process, Ti layer is replaced by a $(Ti_xNb_{1-x})O_2$ compound, which acts as a new diffusion source, and is not perfectly lattice-matched to $LiNbO_3$ [1]. Then, Ti diffusing in substitutional sites, induces important strains in the crystal. From crystallographic analysis on Y-cut samples, it turns out that a linear relationship between induced strain field and Ti concentration cannot completely explain experimental results [2]. In this condition, nonlinear relations depending on C_o are reasonable to consider. Further crystallographic investigations on Z-cut crystals, diffused in a wider range of conditions, should be fruitful to carry on to ascertain the above feelings.

As a consequence, the comparison of $LiNbO_3$ and Ti : $LiNbO_3$ refractive index dispersion, is diffusion dependent. The ordinary and extraordinary index changes at the surface are plotted in Fig. 11 for two typical values of C_o (0.2 and 2 percent). Normalized units are chosen as in [9]. For the considered τ / d_z values, the dispersion of Ti : $LiNbO_3$ is very different from that of the bulk crystal, justifying the goal of the work. The dispersion of extraordinary index change tends to stabilize in the 1.15–1.5 μm region, while this tendency is not present in the ordinary index change behavior. These differences have important consequences on the design of Ti : $LiNbO_3$ waveguides. It appears also, that the influence of diffusion conditions is much stronger on n_e than on n_o dispersion.

VI. Conclusion

Z-cut $LiNbO_3$: Ti planar waveguides have been characterized over a wide range of diffusion conditions : Ti mass concentration varying from 0.2 to 2 percent in surface, and diffusion depth from 4 to 15 μm.

Within measurement error, Ti profiles and induced extraordinary and ordinary index profiles are Gaussian. The dependence of these Gaussian functions with diffusion conditions (τ, t, T) have been established at several wavelengths (0.6328, 1.1523 and, 1.523 μm).

In the assumption of constant diffusion coefficient, the relation between refractive index change $\Delta n_{e,o}(Z)$ and titanium concentration $C(Z)$ can mathematically be expressed as nonlinear and of the form:

$$\Delta n_{e,o}(Z) = A_{e,o}(C_o, \lambda)\left(C(Z)\right)^{\alpha_{e,o}}.$$

The proportionality coefficient $A_{e,o}$ depends not only on the wavelength λ, but also on the diffusion parameters (characterized by C_o : Ti surface concentration).

The dispersion of Ti indiffused $LiNbO_3$ is rather different from that of undoped substrate. The dispersion of extraordinary index change is rather small over the 1.1–1.6 μm wavelength range, whereas it is not true for the ordinary index. These data have been successfully used to model directional couplers characterized at 1.3 and 1.5 μm [29].

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